LAB: 03

This lab focuses on recovering deleted files in FAT system. It begins by covering deleting files in FAT, then moves on to the subject of file recovery tools that forensic investigators use. Then next, you will use FTK manager to make an image file of a USB.

Objective:

- **Learn some basic filesystem analysis and recovery files**

Leaning Activities:

At the end of these activities, you should understand:

- Delete files
- Understand what happens when a file is deleted
- How to recover deleted files
- How to image a storage device

Tools to use:
-part1.001
- part2.001
- USB key
- WinHex
-FTK Imager (http://marketing.accessdata.com/ftkimager4.2.0)
- Disk Editor (https://www.disk-editor.org/index.html)

Note: Please include a screenshot of each step. Corp your screenshot to show relevant information only. Uncropped screenshots will result in a 10% deduction from your marks.

Task 1:

1.1 First download part1.001

1.2 Launch Winhex

1.3 File open "part1.001"

1.4 Identify the signature of "**55 AA**"
At the end of the MBR is the 'magic number' this is signified by **0x55AA**, it shows the end of the MBR area of a disk.

1.5 What is the offset for signature "55 AA": 0x01FE to 0x01FF

1.6 What is the sector number? Sector number for given case is 0. Sector number is number given to region of storage after every 512 bytes, which in this case the bytes 55 AA are still in first 512 bytes so the sector number is 0.

Before **0x55AA,** we can see information for partition entries. These are in 16 byte lengths.
Let's go back 64 bytes before 55AA and see what's there.

On the provided image, we can see a series of hex values which describe the partitions which exist on the disk the image was taken of.

Highlight the preceding 64 bytes before the magic number (the end of the MBR):

You should see **00 20 21 00 06 AD 1E 0F 00 08 00 00 00 D0 03 00**

Yellow – Represents whether the partition is bootable ***(00 means not active)***
Green – This is the starting head 20, cylinder 21 and sector 00 (in little endian) (not important because it limited the theoretical disk sizes)
Blue – Represents the filesystem (important)

1.5 **What filesystem is signified by 06? According to the partition table of MBR, 06 signifies FAT filesystem to be exact FAT16. Hint: Reference (1))**

Red – Address of the end of the partition on the disk
Brown/Green - Logical Block Address of first sector in the partition
Purple- The number of sectors on the drive? (we can calculate the correct drive size based on this value if we know the sector size (512 bytes in this case))

1.6 **What is the size of the drive (MB)? {show your calculations}**
From reference we can see that we can calculate the size of the drive by the number of sectors, which in this case is "00 D0 03 00" from which we can calculate the number of sectors.
00 0D 03 00 is little-endian so normal order is 00 03 D0 00 now using hex to bytes calculator we found out that there are 249,856 sectors, alternatively we can calculate ourselves by
0x0003D000 = (3 x 65536) + (208 x 256) + (0)
= 249,856 sectors
(We get power of 256 from the position of each byte i.e. for 1st position 256 has 0 power which is 1 and then for 2nd position 256 has 1 so 256 and so on)
Now, converting sectors into MB,

MB = (sectors x 512(bytes per sector)) / (1024 x 1024) (first "1024" to convert bytes into bits and second for bits into megabits)
= (249,856 x 512) / 1,048,576
= 127,926,272 / 1,048,576
= 122 MB

Task 2:

Describe the following hex value drive based on the previous example!
80 20 21 00 07 DD 1E 3F 00 08 00 00 00 A0 0F 00

2.1 Is the drive bootable? (yellow**) "80" shows drive is bootable.**

2.2 What filesystem exists on the drive (blue)? From "07" and MBR table, we can say that the filesystem is NTFS.

2.3 How large is the partition (MB)? **{show your calculations}**
From little-endian to normal "00 A0 0F 00"is transformed to 00 0F A0 00. From there using hex to sectors calculator, we get 1,024,000 sectors.
MB = (1024000 x 512) / (1024 x 1024)
= (512000)/1024
= 500 MB

Task 3:

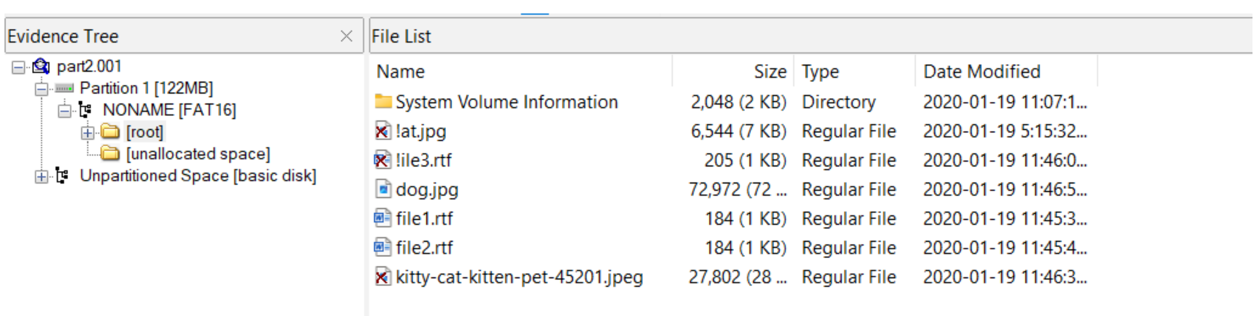
3.1 Download part1.002 image

3.2 Lunch FTK imager

3.3 File, "Add Evidence Item"

3.4 Select "Image File" and select part2.001

3.5 When the image loads, in the Evidence Tree pane open Partition 1, NONAME [FAT 16], root.



3.6 Now in the 'File List' you will see 3 files which look normal and the other 3 look like deleted files as indicated in Figure 2.6.
However, for the files which have been deleted, the first letter of the file name has been changed from the correct letter to an E5 (hex value). This is to signify in the file allocation table that the file is no longer present and that it can be over-written.

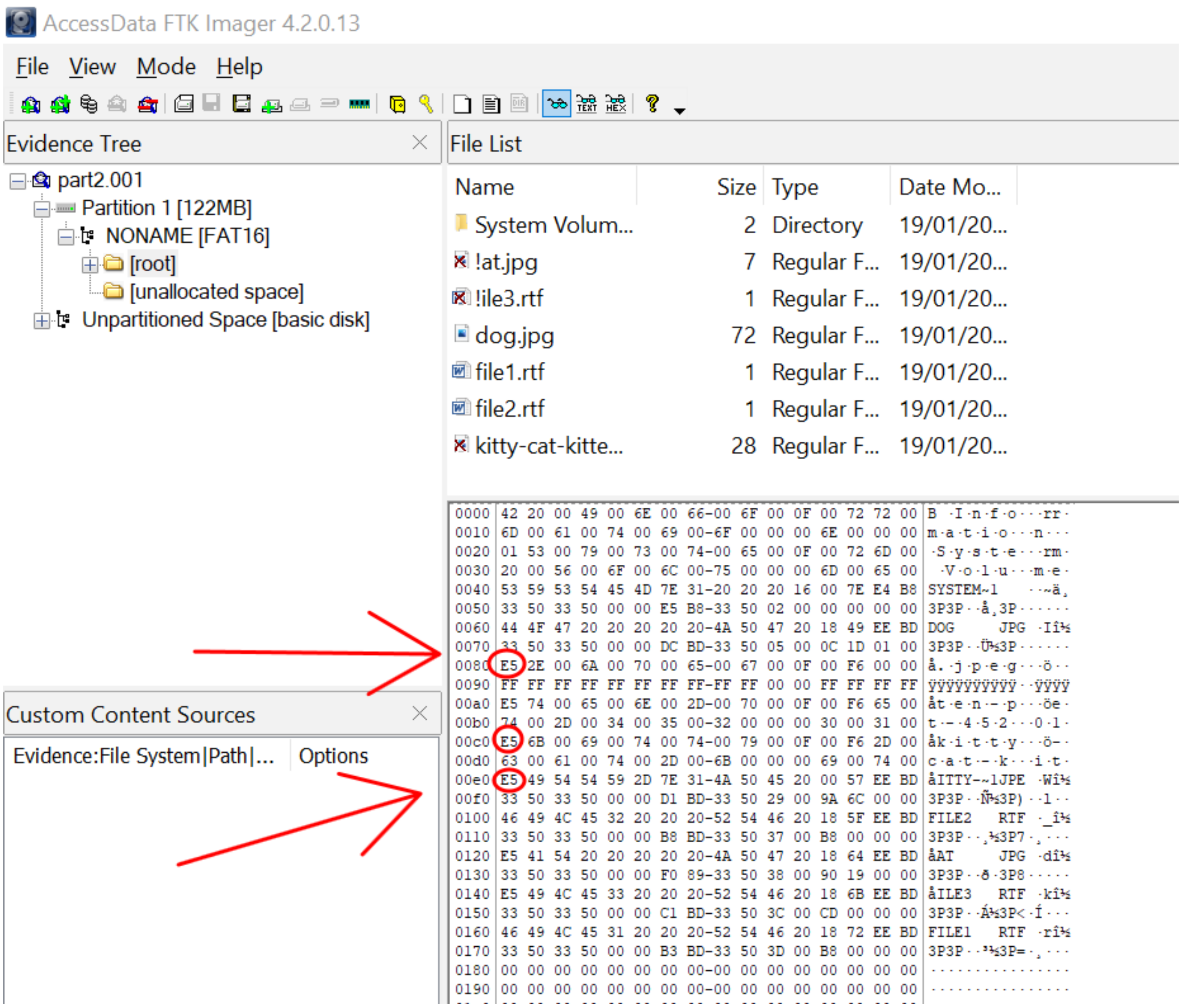
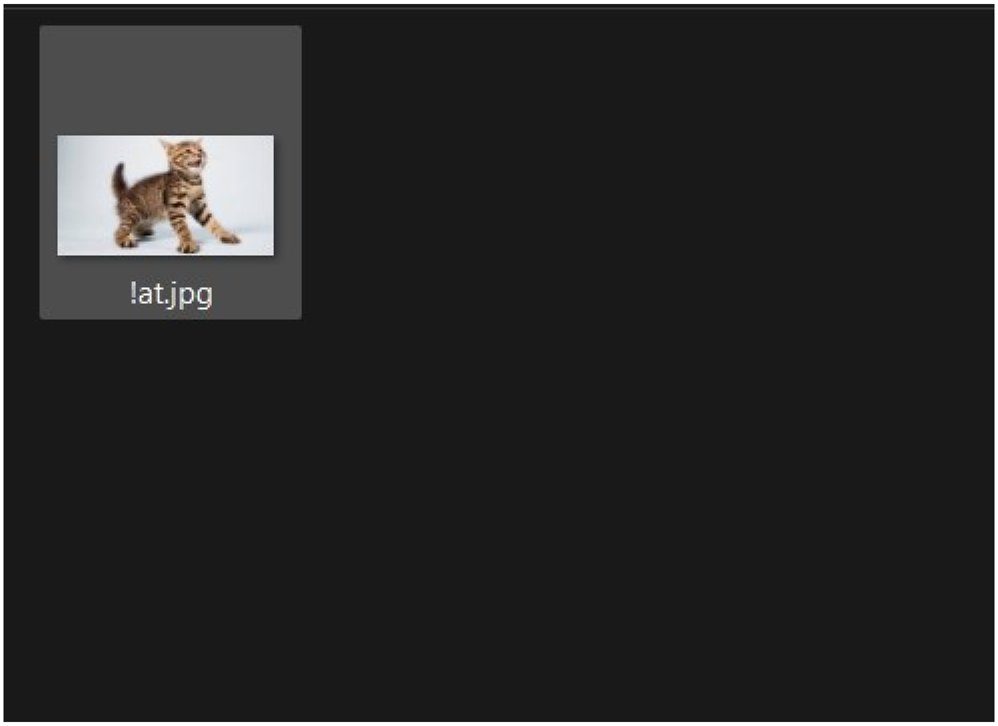


Figure 2.6

3.7 Right click on "!at.jpg" file, Export Files, Browse For Folder to recover the deleted file.



Task 4:

4.1 Let’s add a file to a USB keys, then delete the file off of your USB.

4.2 Now, run WinHex as administrator

4.3 Select Tools, Open Disk then select your usb disk that you just deleted the file from under ‘physical storage devices’

4.4 In the top Pane, click on ‘Root directory’ (Figure 3.4):

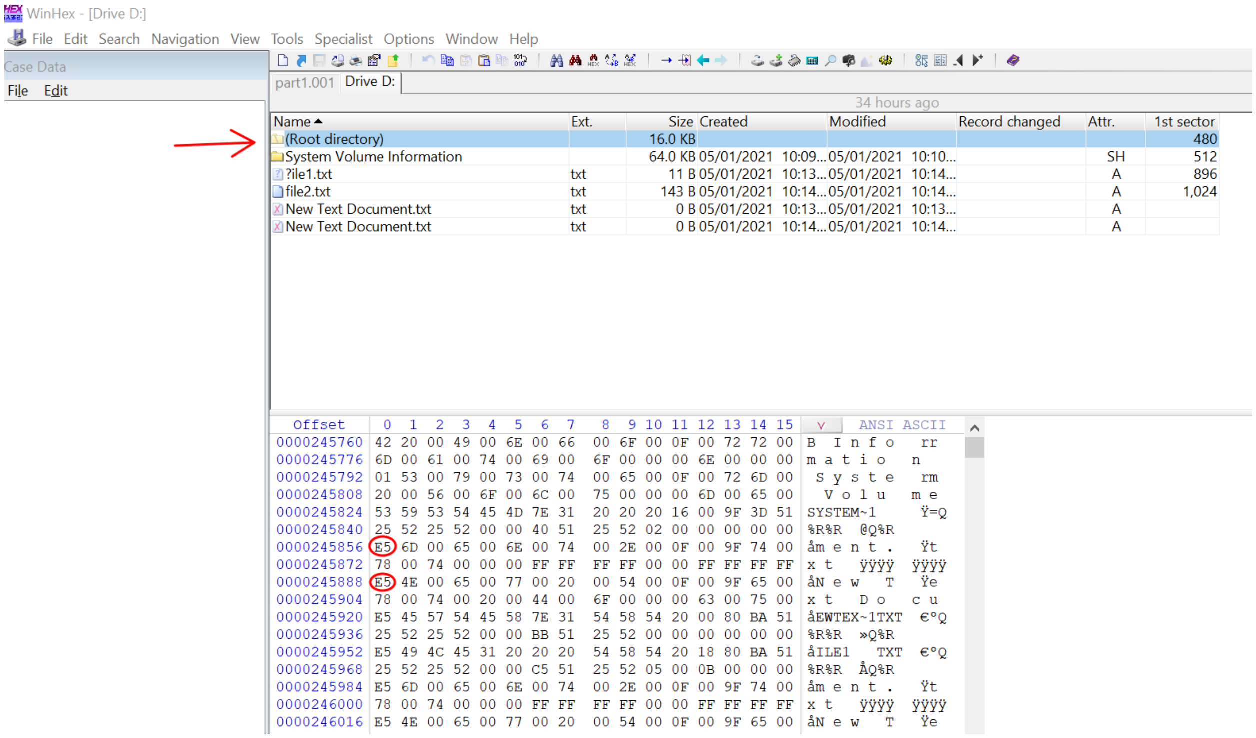
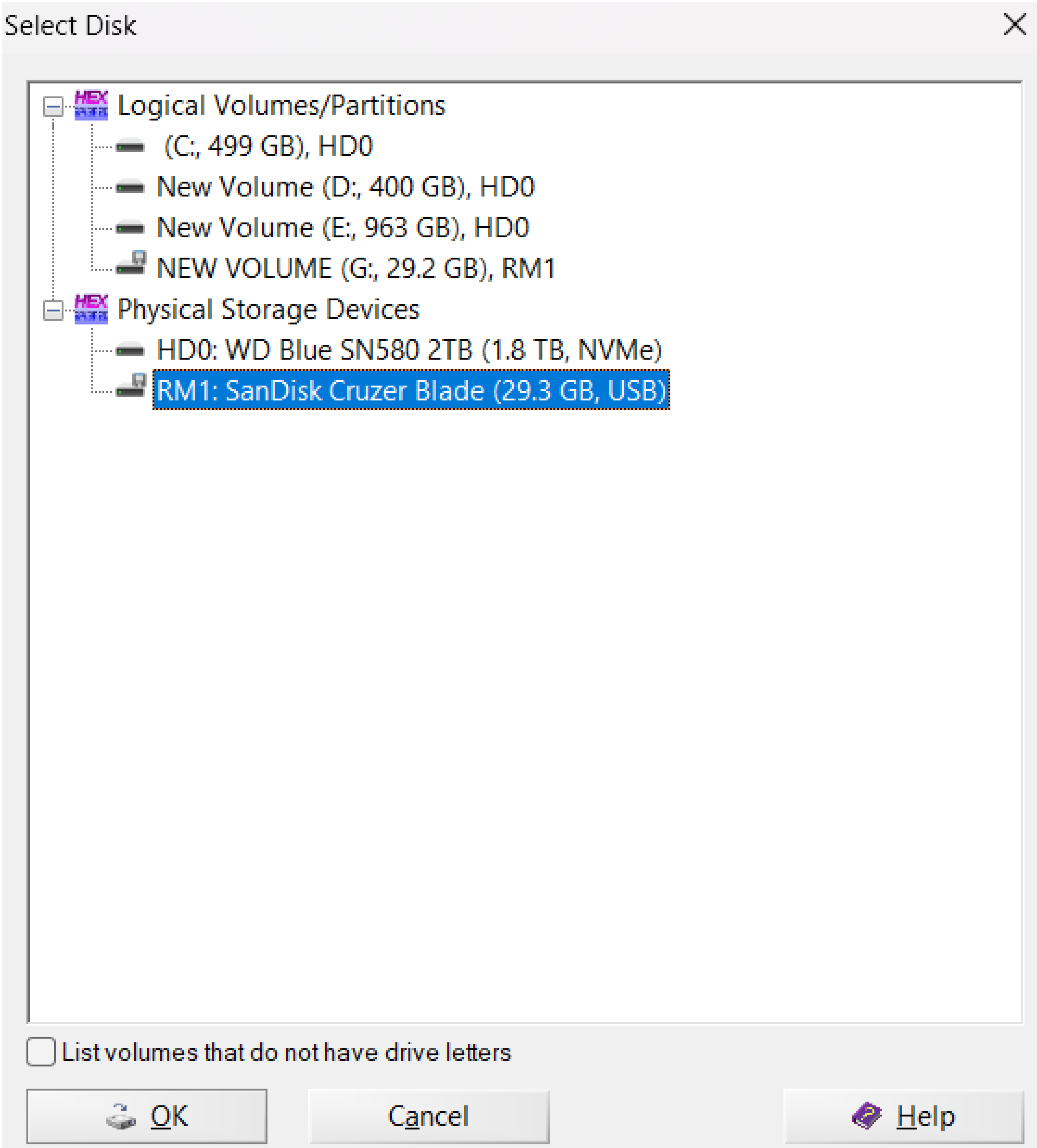


Figure 3.4

In the lower hex view, you should see your file's filename with the E5 character for it's first letter. With the free version of winhex we can't change this value, if winhex would allow us, we could 'undelete' this file by removing the E5 flag from the filename.

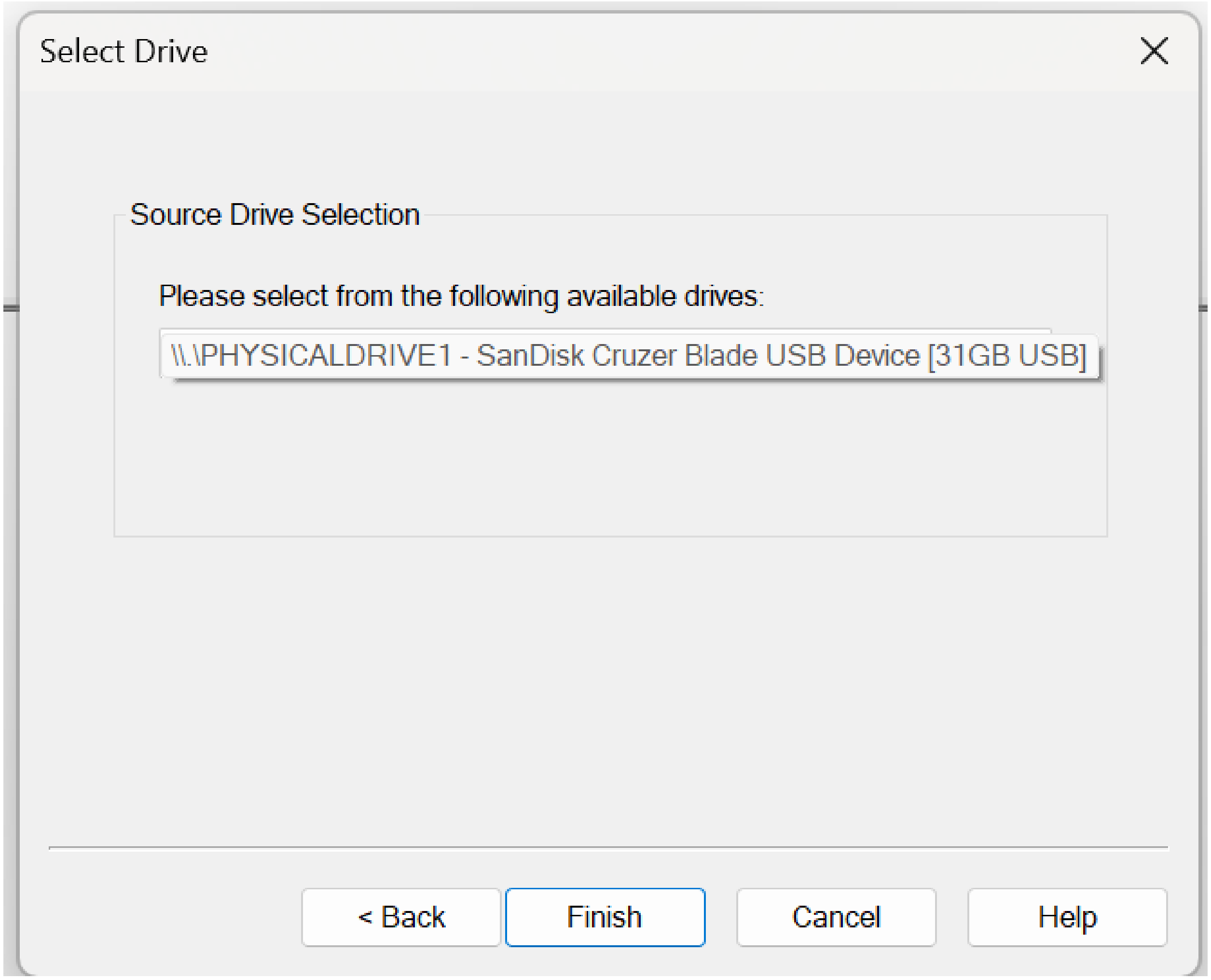


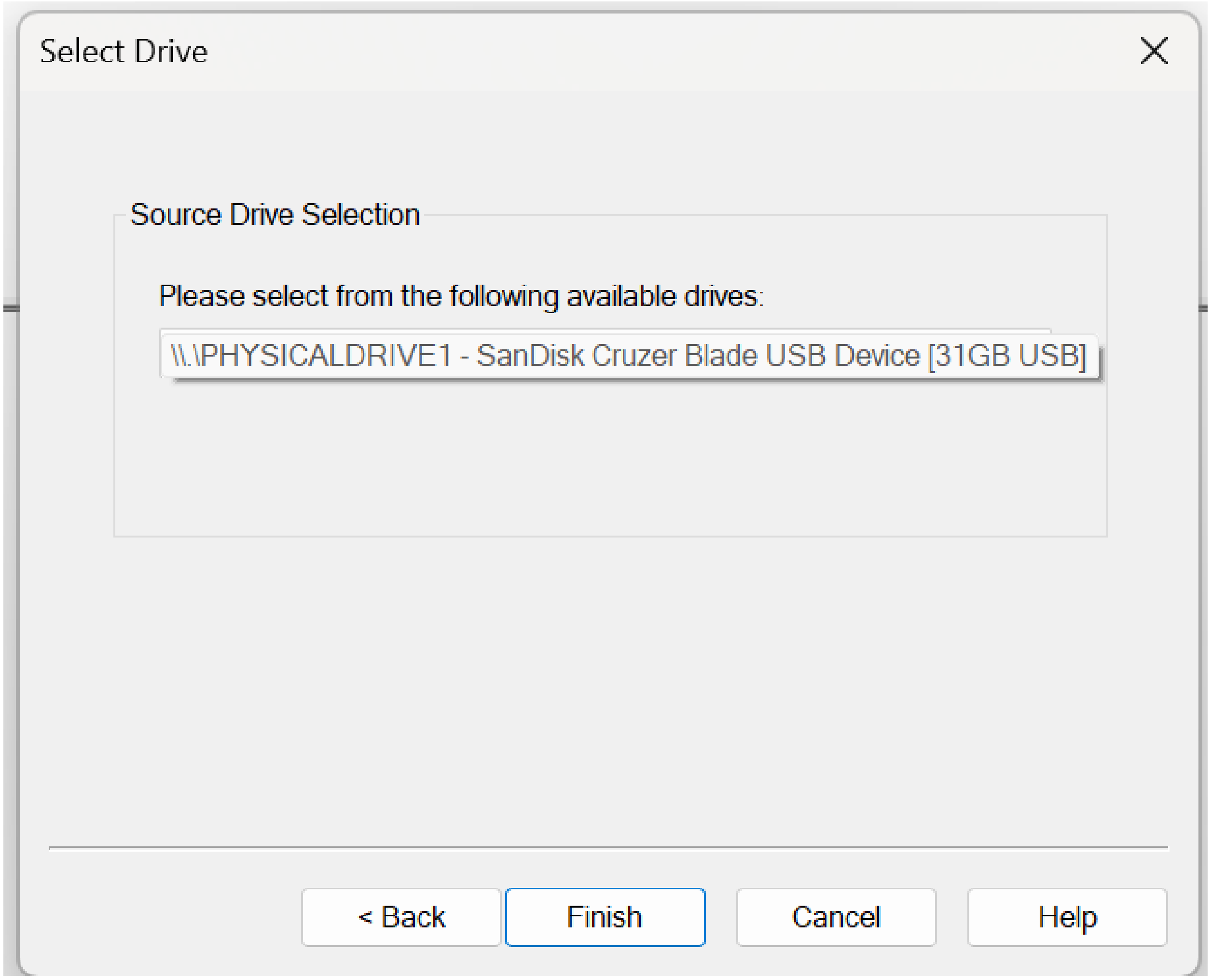
Name	Ext.	Size	Created	Modified	Record changed	Attr.	1st sector
(Root directory)		16.0 KB					32,768
System Volume Information		16.0 KB	2025-05-30 16:16:...	2025-05-30 16:16:...		SH	32,800
New Text Document.txt	txt	0 B	2025-05-30 16:21:...	2025-05-30 16:21:...		A	
Test 1.txt	txt	11 B	2025-05-30 16:21:...	2025-05-30 16:22:...		A	32,864

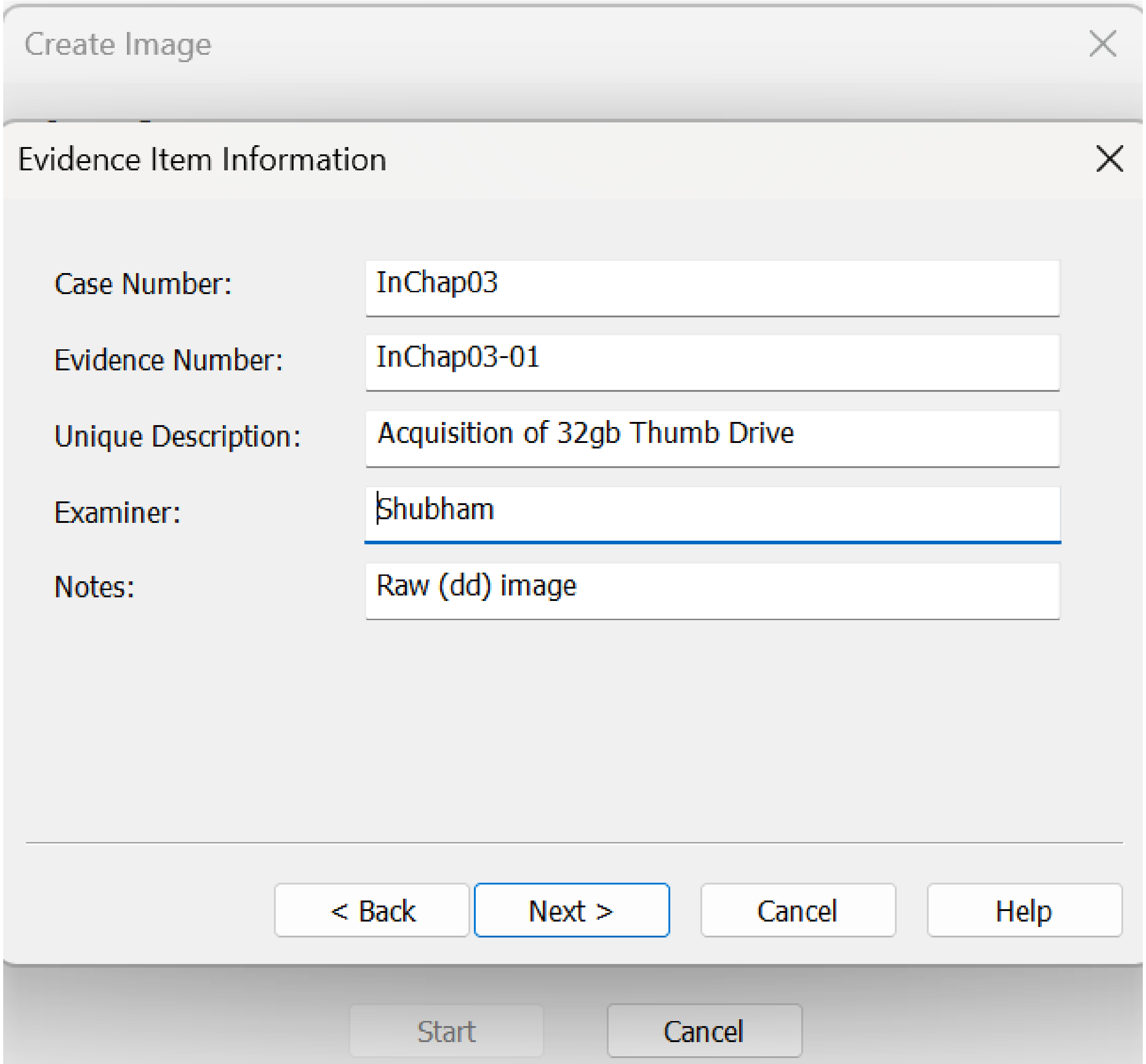
Task 5:

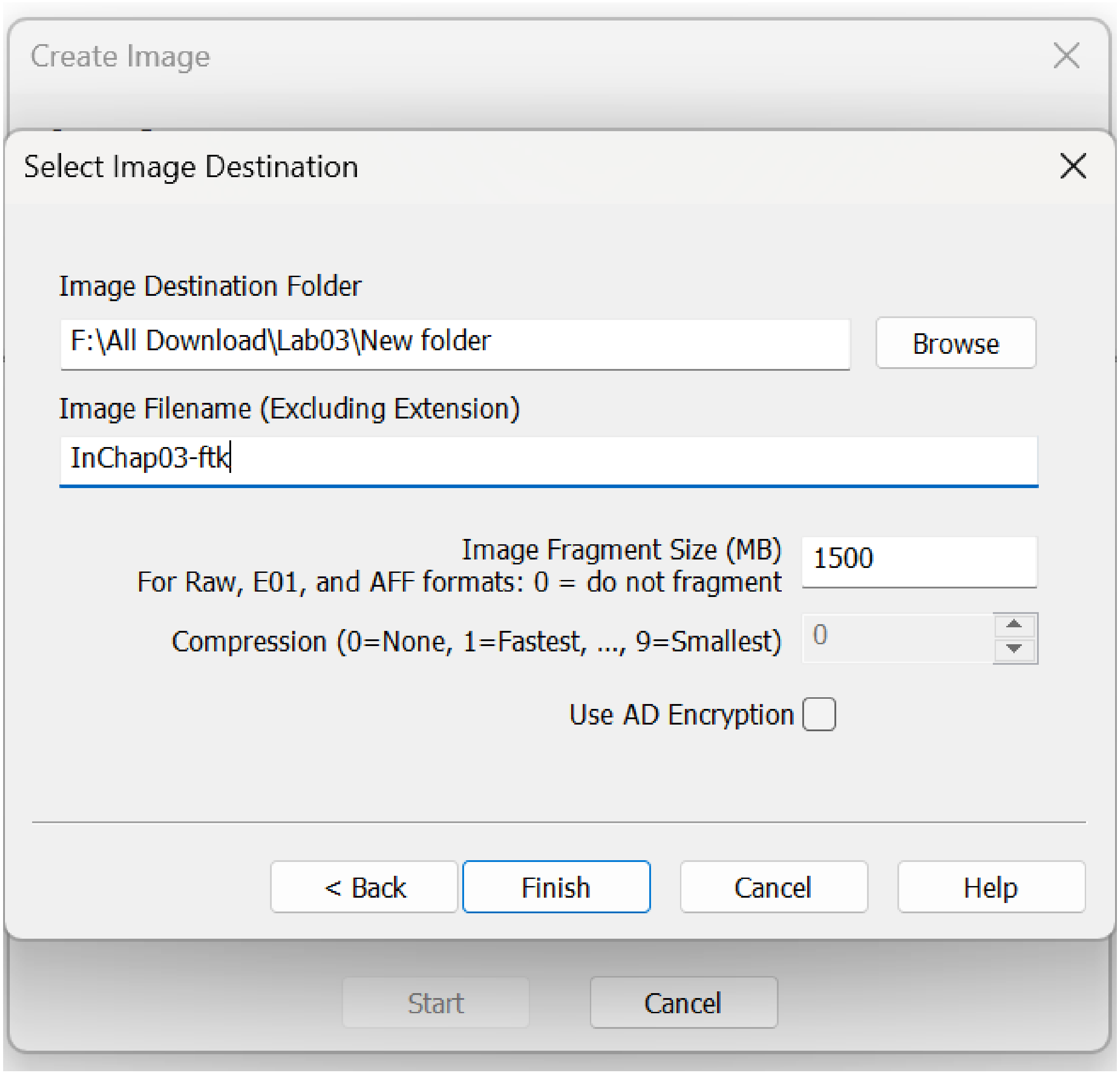
Text book, Page 118 & 121

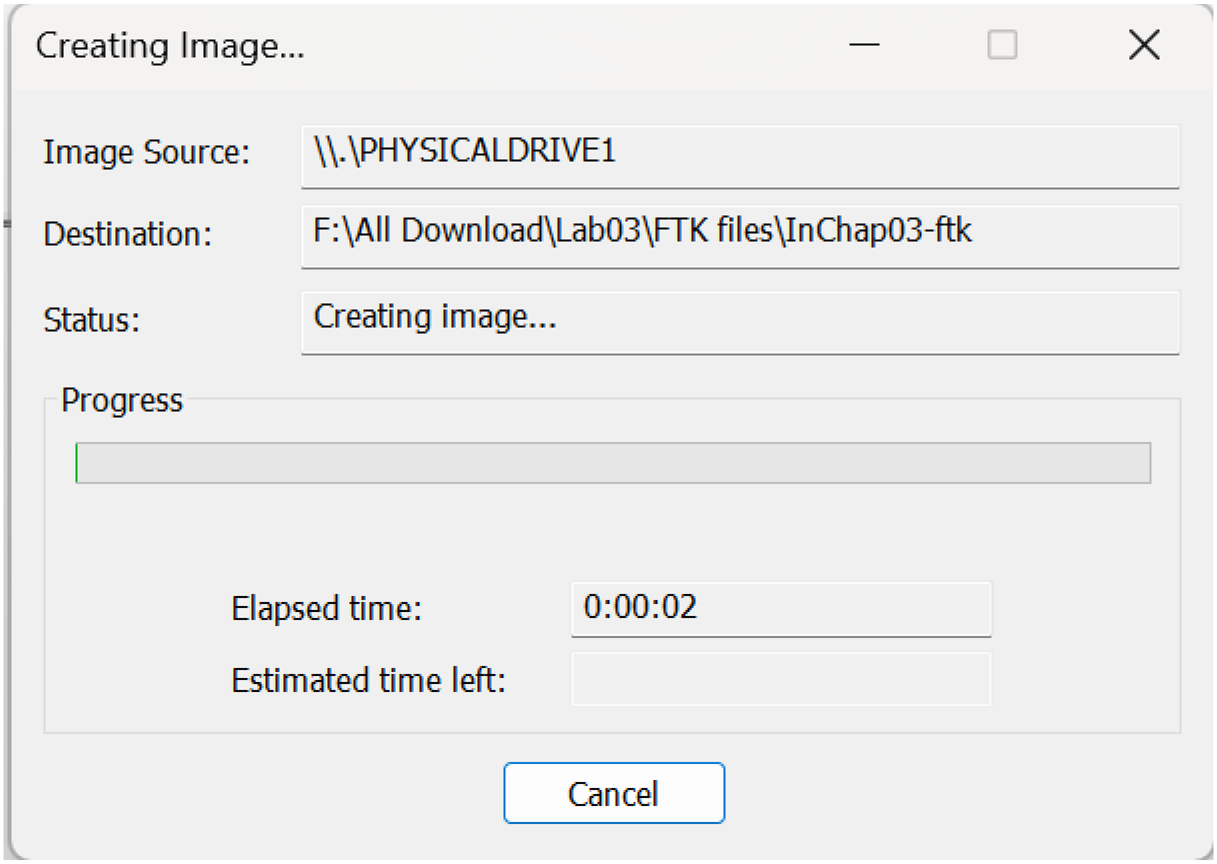
Steps 1-14


Select Drive
Source Drive Selection
Please select from the following available drives:
\\.\PHYSICALDRIVE1 - SanDisk Cruzer Blade USB Device [31GB USB]
< Back
Finish
Cancel
Help


Select Drive
Source Drive Selection
Please select from the following available drives:
\\.\PHYSICALDRIVE1 - SanDisk Cruzer Blade USB Device [31GB USB]
< Back
Finish
Cancel
Help

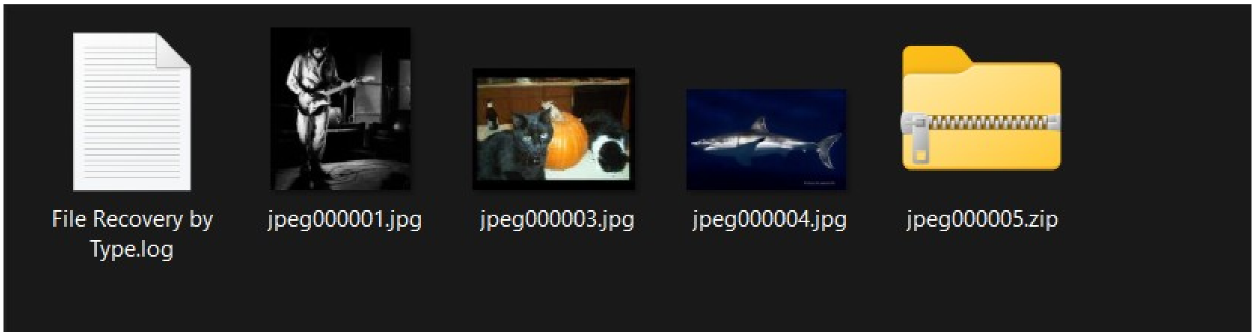

Create Image
Evidence Item Information
Case Number:
InChap03
Evidence Number:
InChap03-01
Unique Description:
Acquisition of 32gb Thumb Drive
Examiner:
Shubham
Notes:
Raw (dd) image
< Back
Next >
Cancel
Help
Start
Cancel


Create Image
Select Image Destination
Image Destination Folder
F:\All Download\Lab03\New folder
Browse
Image Filename (Excluding Extension)
InChap03-ftk
Image Fragment Size (MB)
For Raw, E01, and AFF formats: 0 = do not fragment
1500
Compression (0=None, 1=Fastest, ..., 9=Smallest)
0
Use AD Encryption
< Back
Finish
Cancel
Help
Start
Cancel



Task 6:

6.1 Open Winhex
6.2 File Open and select lab3.dd
6.3 Tools, Disk Tools, File Recover by Type
6.4 Select JPEG and Zip Archive in "File Header Search on lab3.dd"
6.5 Select "Output patch" to C:\lab3\task6
6.6 Verify the files were recovered in the above folder
6.7 How many file files were recovered? 4 files and 1 log file



6.8 What are the file names in step7?
File Recovered by Type.log, jpeg000001.jpg, jpeg000003.jpg, jpeg000004.jpg, jpeg000005.zip

Task 7:

MFT Little Endian Format
Convert the following from Big Endian to Little Endian Format to Decimal Format:

7.1 60 00: Little endian = 00 60, Decimal Format = 96
7.2 80 00: Little endian = 00 80, Decimal Format = 128
7.3 00 40 00 20: Little endian = 20 00 40 00, Decimal Format = 536,879,104
7.4 30 00 10 20: Little endian = 20 10 00 30, Decimal Format = 537,001,392
7.5 00 A0 0F 00: Little endian = 00 0F A0 00, Decimal Format = 1,024,000

Reflective statements (end-of-exercise):
You should reflect on these questions:

1. What are BitLocker's current hardware and software requirements?

Hardware requirements are: TPM version 1.2 or 2.0, TCG-compliant BIOS/UEFI, and must have at least 2 partitions, Windows 11/10/8.1/8/7 (not available in Home version), NTFS filesystem.

2. What is a 'Root Directory'?

   Root directory is directory where the main operating system files are saved. It is most important part for OS since all the files required to run the OS are saved there

3. Briefly describe how delete command function in FAT16 version?

   When a file is deleted under FAT16, it is not instantly removed from the disk. Rather, the space the file took up is designated as usable again, and only the directory entry is changed.
   Step 1: First byte of filename → E5 (marks it as deleted)
   Step 2: FAT cluster chain cleared (space marked as free)
   Step 3: File data remains until overwritten

References

Data Recovery. (n.d.). *Hexadecimal flags for partition type*. Retrieved May 30, 2025, from https://datarecovery.com/rd/hexadecimal-flags-for-partition-type/
X-Ways Software Technology AG. (n.d.). *WinHex: Computer forensics & data recovery*. Retrieved May 30, 2025, from https://www.x-ways.net/winhex/
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Microsoft. (2023, March 15). *Description of FAT, HPFS, and NTFS file systems*. Microsoft Support. https://docs.microsoft.com/en-us/troubleshoot/windows-client/backup-and-storage/fat-hpfs-and-ntfs-file-systems
A.E. Brouwer. (2000). *Partition types*. Eindhoven University of Technology. https://www.win.tue.nl/~aeb/partitions/partition_types-1.html